## Laplace Transformu

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$x(t) \xleftrightarrow{\mathcal{L}} X(s)$$

$$X(s) = \mathcal{L}\{x(t)\}$$

$\Rightarrow s = \sigma + j\omega$ $\longrightarrow$ $\sigma$: Reel kısım

$\hookrightarrow$ $\omega$: Sanal kısım

$$X(\sigma+j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-(\sigma+j\omega)t} dt = \int_{-\infty}^{\infty} x(t)\, e^{-\sigma t}\, e^{-j\omega t} dt = \mathcal{F}\{x(t)e^{-\sigma t}\}$$

$\Rightarrow s = j\omega$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t} dt = \mathcal{F}\{x(t)\} = X(s)\Big|_{s=j\omega}$$

Laplace transformu, $x(t)e^{-\sigma t}$ şeklinde $x(t)$ işaretinin bir reel üstel işaret ile çarpımının Fourier transformudur.

$e^{-\sigma t}$ $\longrightarrow$ $\sigma > 0$ zamanla azalan fonksiyon

$\hookrightarrow$ $\sigma < 0$ zamanla artan fonksiyon

Ex: $x(t) = e^{-at}u(t)$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} e^{-at}u(t) e^{-st} dt = \int_{0}^{\infty} e^{-(a+s)t} dt = \frac{e^{-(a+s)t}}{-(a+s)}\Big|_0^{\infty} = \frac{1}{s+a}, \quad Re\{s+a\} > 0$$

$$s = \sigma + j\omega \longrightarrow X(\sigma+j\omega) = \frac{1}{(\sigma+a)+j\omega}, \quad \sigma + a > 0$$

$$s = j\omega, \ \sigma = 0 \longrightarrow X(j\omega) = \frac{1}{a+j\omega}, \quad a > 0$$

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}, \quad Re\{s\} > -a \quad \Big\downarrow a=0$$

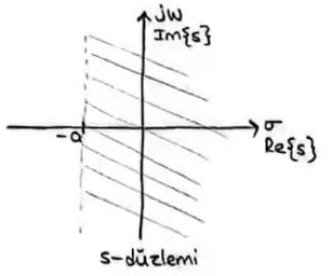
$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}, \quad Re\{s\} > 0$$

Ex: $x(t) = -e^{-at}u(-t)$

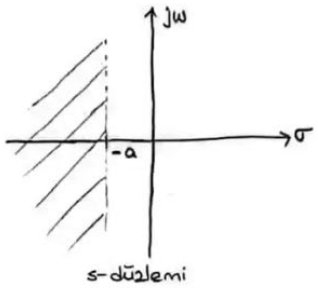
$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = -\int_{-\infty}^{\infty} e^{-at}u(-t)\, e^{-st} dt = -\int_{-\infty}^{0} e^{-(a+s)t} dt = \frac{1}{s+a}, \quad Re\{s+a\} < 0$$

$$-e^{-at}u(-t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}, \quad Re\{s\} < -a$$

⇒ Yakınsaklık bölgesi (Region of convergence, ROC): Laplace transform integralının yakınsadığı s değer aralığına yakınsaklık bölgesi denir.



s-düzlemi



s-düzlemi

$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$, $Re\{s\} > -a$

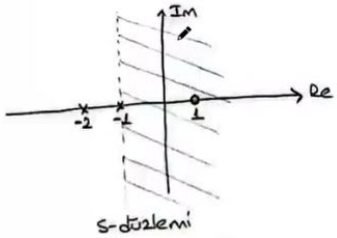
$\mathcal{L}\{-e^{-at}u(-t)\} = \frac{1}{s+a}$, $Re\{s\} < -a$

Ex: $x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$

$e^{-2t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+2}$, $Re\{s\} > -2$

$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1}$, $Re\{s\} > -1$

$X(s) = \frac{3}{s+2} - \frac{2}{s+1} = \frac{s-1}{s^2+3s+2}$, $Re\{s\} > -1$



s-düzlemi

kutup
x: pole (root of denumerator)

o: zero (root of numerator)
sıfır

Ex: $x(t) = \delta(t) - \frac{4}{3}e^{-t}u(t) + \frac{1}{2}e^{2t}u(t)$

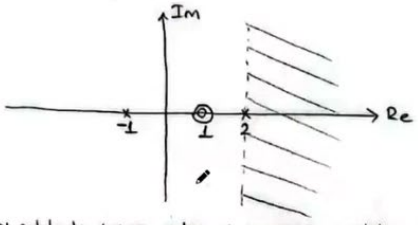
$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(t)e^{-st}dt = \int_{-\infty}^{\infty} \delta(t)e^{-s0}dt = \int_{-\infty}^{\infty} \delta(t)dt = 1$

$\delta(t) \xleftrightarrow{\mathcal{L}} 1$, $\forall s$ (ROC, bütün s-düzlemi)

$e^{-t}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+1}$, $Re\{s\} > -1$

$e^{2t}u(t) \longleftrightarrow \frac{1}{s-2}$, $Re\{s\} > 2$

$X(s) = 1 - \frac{4}{3}\frac{1}{s+1} + \frac{1}{2}\frac{1}{s-2} = \frac{(s-1)^2}{(s+1)(s-2)}$, $Re\{s\} > 2$




⇒ Aynı noktada tekrar eden s sayısına mertebe denir.

Paydada, $s=-1$ ve $s=2$ olmak üzere birinci mertebeden iki adet kutup ve payda $s=1$ 'de ikinci mertebeden bir adet sıfır vardır.

Not: Laplace transformunun ROC'u jw eksenini içermiyorsa Fourier transformu yakınsamaz.

$\frac{1}{3}e^{2t}u(t)$ ifadesinin Fourier transformu yoktur.

Yakınsama bölgesi (ROC) özellikleri:

1) $X(s)$'in ROC'u s-düzlemindeki jw eksenine paralel şeritlerden oluşur.

$$X(s) = \underset{s=\sigma+j\omega}{\uparrow} X(\sigma+j\omega) = \int_{-\infty}^{\infty} x(t)e^{-st}dt = \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}dt = \mathcal{F}\{x(t)e^{-\sigma t}\}$$

$$\int_{-\infty}^{\infty} |x(t)|e^{-\sigma t}dt < \infty$$

2) Rasyonel Laplace transform için, ROC herhangi bir kutup içermez.

3) $x(t)$ sonlu süreli ve mutlak entegre edilebilir ise, ROC bütün s düzlemidir.

$$\int_{T_1}^{T_2} |x(t)|dt < \infty$$

$$\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}dt < \infty \leftarrow \text{1) özellik}$$

→ $\mathrm{Re}\{s\}=\sigma=0$ iken $s=j\omega$ olup ROC içinde kalır.

→ $\mathrm{Re}\{s\}=\sigma>0$ iken $\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}dt < e^{-\sigma T_1}\underbrace{\int_{T_1}^{T_2} |x(t)|dt}_{\infty} < \infty$ olur ve $\mathrm{Re}\{s\}>0$ sağ düzlemi ROC içinde kalır.

→ $\mathrm{Re}\{s\}=\sigma<0$ iken $\int_{T_1}^{T_2} |x(t)|e^{-\sigma t}dt < e^{-\sigma T_2}\int_{T_1}^{T_2} |x(t)|dt < \infty$ olur ve $\mathrm{Re}\{s\}<0$ sağ düzlemi ROC içinde kalır.

Ex: $x(t) = \begin{cases} e^{-at}, & 0<t<T \\ 0, & \text{dışında} \end{cases}$

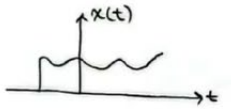
$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt = \int_{0}^{T} e^{-(a+s)t}dt = \frac{1}{s+a}\left(1-e^{-(a+s)T}\right)$$

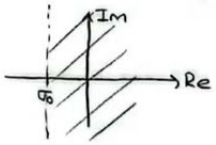
← 3)'e göre ROC bütün s düzlemi
2)'e göre ROC kutup içermez, $s=-a$ kutup var.

$s=-a$ iken $X(s) = \frac{0}{0}$

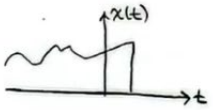
$$\lim_{s\to -a} X(s) = \lim_{s\to -a}\left(\frac{\frac{d}{ds}\left(1-e^{-(a+s)T}\right)}{\frac{d}{ds}(s+a)}\right) = \lim_{s\to -a} Te^{-aT}e^{-sT} \Longrightarrow X(-a)=T$$

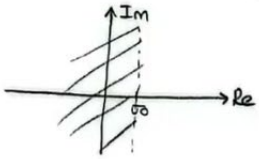
4) $x(t)$ sağ yanlı ve $Re\{s\}=\sigma_0$ doğrusu ROC'da ise, $Re\{s\}>\sigma_0$ için bütün s değerleri de ROC'da olur.



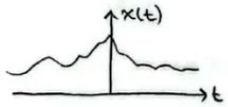



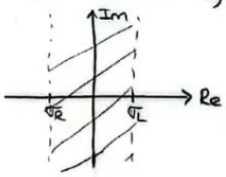

5) $x(t)$ sol yanlı ve $Re\{s\}=\sigma_0$ doğrusu ROC'da ise, $Re\{s\}<\sigma_0$ için bütün s değerleri de ROC'da olur.




6) $x(t)$ çift yanlı ve $Re\{s\}=\sigma_0$ doğrusu ROC'da ise, ROC, s düzleminde $Re\{s\}=\sigma_0$ doğrusunu içeren bir şeritten oluşur.




7) $X(s)$ rasyonel ise, ROC, ya kutuplarla sınırlıdır ya da sonsuza genişler ve kutup içermez.

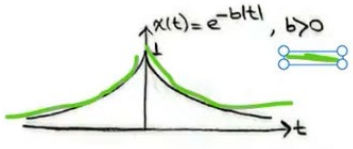
→ $x(t)$ sağ yanlı ise, ROC, en sağdaki kutubun sağında kalan s düzlemidir.

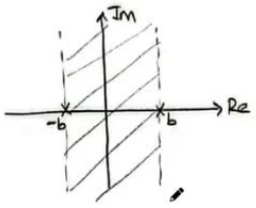
→ $x(t)$ sol yanlı ise, ROC en soldaki kutubun solunda kalan s düzlemidir.

Ex: $x(t)=e^{-b|t|}$

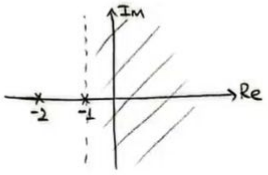
$$X(s)=\int_{-\infty}^{\infty} x(t)e^{-st}dt = \int_{-\infty}^{0} e^{bt}e^{-st}dt + \int_{0}^{\infty} e^{-bt}e^{-st}dt = \frac{-1}{s-b}+\frac{1}{s+b}, \quad \begin{matrix} Re\{s-b\}<0 \\ Re\{s+b\}>0 \end{matrix}$$

$$X(s)=\frac{-2b}{s^2-b^2}, \quad -b<Re\{s\}<b$$



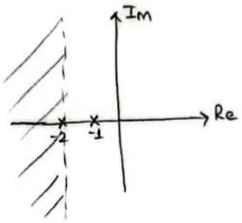
Ex: $X(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$




x(t) sağ yanlı ise

$x(t) = e^{-t}u(t) - e^{-2t}u(t)$

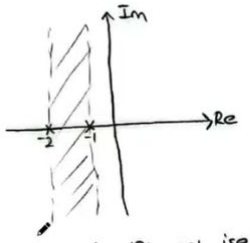
ROC, $s=j\omega$ içerir, FT yakınsar.




x(t) sol yanlı ise

$x(t) = -e^{-t}u(-t) + e^{-2t}u(-t)$

FT yoktur




x(t) çift yanlı ise

$x(t) = -e^{-t}u(-t) - e^{-2t}u(t)$

FT yoktur.

## Ters Laplace Transform:

$$X(s) = X(\sigma+j\omega) = \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\omega t}dt = \mathcal{F}\{x(t)e^{-\sigma t}\}$$

$$x(t)e^{-\sigma t} = \mathcal{F}^{-1}\{X(\sigma+j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma+j\omega)e^{j\omega t}d\omega$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma+j\omega)e^{(\sigma+j\omega)t}d\omega$$

$s = \sigma+j\omega \longrightarrow ds = j\,d\omega$

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}ds$$

$$\Rightarrow X(s) = \sum_{i=1}^{n}\frac{A_i}{s+a_i} \longrightarrow x(t) = A_i e^{-a_i t}u(t) : \text{sağ yanlı}$$

$$\longrightarrow x(t) = -A_i e^{-a_i t}u(-t) : \text{sol yanlı}$$

## Laplace Transformının Özellikleri

$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s)$, ROC : R

$X(s) = \mathcal{L}\{x(t)\}$ $\quad x(t) = \mathcal{L}^{-1}\{X(s)\}$

1) Lineerlik: $\alpha x_1(t) + \beta x_2(t) \longleftrightarrow \alpha X_1(s) + \beta X_2(s)$, ROC : $R_1 \cap R_2$

2) Zamanda öteleme: $x(t-t_0) \longleftrightarrow e^{-st_0}X(s)$, ROC : R

3) s-düzleminde öteleme: $e^{s_0 t}x(t) \longleftrightarrow X(s-s_0)$, ROC : $R+Re\{s_0\}$

$s_0 = j\omega_0 \Rightarrow e^{j\omega_0 t}x(t) \longleftrightarrow X(s-j\omega_0)$, ROC : R

4) Zamanda ölçeklendirme: $x(at) \longleftrightarrow \frac{1}{|a|}X(s/a)$, ROC : R/a

$a=-1 \Rightarrow x(-t) = X(-s)$, ROC : -R

5) Eşlenik alma: $x^*(t) \longleftrightarrow X^*(s^*)$, ROC : R

6) Türev: $\frac{dx(t)}{dt} \longleftrightarrow sX(s)$, ROC : R'yi içerir.

$-tx(t) \longleftrightarrow \frac{dX(s)}{ds}$, ROC : R

7) Integral: $\int_{-\infty}^{t} x(\tau)d\tau \longleftrightarrow \frac{1}{s}X(s)$, ROC: $R \cap \{Re\{s\} > 0\}$

8) Konvolüsyon: $x_1(t) * x_2(t) \longleftrightarrow X_1(s)X_2(s)$, ROC: $R_1 \cap R_2$




$y(t) = x(t) * h(t) \longleftrightarrow Y(s) = X(s)H(s)$

$H(s) = \frac{Y(s)}{X(s)}$ : Transfer fonksiyonu

Ex: $x(t) = te^{-at}u(t)$, $X(s) = ?$

$te^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$, $Re\{s\} > -a$

$te^{-at}u(t) \longleftrightarrow -\frac{d}{ds}\left(\frac{1}{s+a}\right) = \frac{1}{(s+a)^2}$, $Re\{s\} > -a$

$\frac{t^2}{2}e^{-at}u(t) \longleftrightarrow \frac{1}{(s+a)^3}$, $Re\{s\} > -a$

$\vdots$

$\frac{t^{n-1}}{(n-1)!}e^{-at}u(t) \longleftrightarrow \frac{1}{(s+a)^n}$, $Re\{s\} > -a$

Ex: $H(s) = \frac{2s^2+5s+5}{(s+1)^2(s+2)}$, $Re\{s\} > -1$, $h(t) = ?$

$H(s) = \frac{A}{(s+1)^2} + \frac{B}{(s+1)} + \frac{C}{(s+2)} \longrightarrow A = 2,\ B = -1,\ C = 3$

$\updownarrow \mathcal{L}^{-1}$

$h(t) = 2te^{-t}u(t) - e^{-t}u(t) + 3e^{-2t}u(t)$

ROC ile LTI sistemlerin özelliklerinin analizi

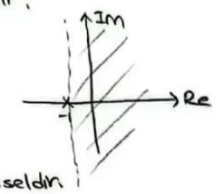
1) Nedensellik: $h(t) = 0$, $t < 0$ sağ yanlı $\longrightarrow$ ROC sağ yanlı

2) Kararlılık: $\int_{-\infty}^{\infty} |h(t)|dt < \infty \longrightarrow$ ROC, jw eksenini içermelidir.

Ex: $h(t) = e^{-t}u(t)$ , kararlı mıdır?

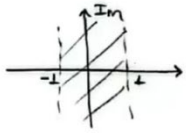
$\updownarrow$

$H(s) = \frac{1}{s+1}$ , $Re\{s\} > -1$



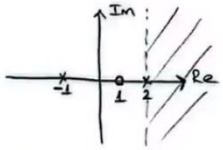

ROC sağ yanlı olduğundan nedenseldir.

Ex: $h(t) = e^{-|t|} \longleftrightarrow H(s) = \frac{-2}{s^2-1}$ , $-1 < Re\{s\} < 1$




Nedensel değildir.

Ex: $H(s) = \frac{s-1}{(s+1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-2}$ , $A = 2/3$, $B = \frac{1}{3}$

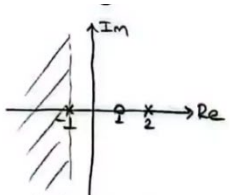



h(t) sağ yanlı

$h(t) = \frac{2}{3}e^{-t}u(t) + \frac{1}{3}e^{2t}u(t)$

Nedensel (sağ yanlı)

Kararlı değil (jw eksenini içermez)

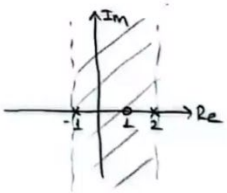



h(t) sol yanlı

$h(t) = -\frac{2}{3}e^{-t}u(-t) - \frac{1}{3}e^{2t}u(-t)$

Nedensel değil (sol yanlı)

Kararlı değil




h(t) çift yanlı

$h(t) = \frac{2}{3}e^{-t}u(t) - \frac{1}{3}e^{2t}u(-t)$

Nedensel değil

Kararlı

Ex: $\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t)$ , $h(t) = ?$

$\updownarrow \mathcal{L}$

$s^2Y(s) + 3sY(s) + 2Y(s) = sX(s) + 3X(s)$

$H(s) = \frac{Y(s)}{X(s)} = \frac{s+3}{s^2+3s+2} = \frac{A}{s+1} + \frac{B}{s+2}$ , $A = 2$, $B = -1$

$\updownarrow \mathcal{L}^{-1}$

$h(t) = 2e^{-t}u(t) - e^{-2t}u(t)$ , $Re\{s\} > -1$